Data Handling & Regression – Assignment 3 – “Real” data

Perform the assignment given below in either MATLAB or Python. Upload in Portflow your code (in the formats .m, .mlx, .py, .ipynb) and a concise report stating your methods, results and conclusions (in .pdf-format, or in the .mlx- or .ipynb-file). If possible, upload your dataset as well (preferably in .csv).

1. Find or construct a dataset yourself based on physical models. Look for inspiration in your project, other courses, or ask students from the pre-grad phase whether they have any data you can analyze for them. Use this as an opportunity to work with data which is not specifically constructed for data analysis.

- The dataset you use must have at least 100 observations of at least 3 explanatory variables and 1 outcome variable. The explanatory variables can be binary, discrete or continuous, but the outcome variable must be binary or discrete.

- It is not allowed to use a dataset from Kaggle or any other freely available source.

- It is also not allowed to artificially create a dataset where the outcome variable is a given function of the explanatory variables.

2. Set up a research question you want to answer with your analysis. If you use data from a pre-grad project, this is probably given to you.

3. Follow a clear, structured path to answer the research question. Explain what steps you have taken and why.

4. Do you have any suggestions to improve the dataset?